

1) Medical Image Enhancement

An x-ray image shows bones clearly, but the darker soft tissues have poor visibility.

a) log Transformation.

b) log Transformation expands the intensity values

in the dark regions of the image

- It enhances low intensity pixels compressing high intensity pixels.

c) • Soft tissues become brighter & easier to observe.

- Hidden details in dark areas are revealed.

2) Night Time Surveillance.

a) log Transformation.

b) log Transformation increases the brightness of low intensity pixels, dark objects visible.

- Therefore, log Transformation provides better visibility for night-time images.

c) • The parking lot appears brighter.

- Overall image contrast in dark region improved.

3) Fingerprint Recognition.

a) power-law (Gamma) Transformation.

b) Gamma Transformation enhances enhanced image contrast by adjusting brightness levels.

- with $\gamma < 1$, brighten & more distinguishable.

c) log Transformation would brighten the entire dark region.

- Fine ridge details may become less distinct reducing recognition accuracy.